

## Work Experience

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| <b>Simula Metropolitan Center for Digital Engineering (SimulaMet)</b><br><i>PhD Student, Affiliated to Oslo Metropolitan University</i> <ul style="list-style-type: none"><li>Research on the potential applications of multimodal understanding across multiple domains.</li></ul>                                                                                                            | <b>Oslo, Norway</b><br>May 2023 – Present      |
| <b>The Alan Turing Institute</b><br><i>PhD Enrichment Student</i> <ul style="list-style-type: none"><li>Student Representative for the 2024/25 cohort.</li></ul>                                                                                                                                                                                                                               | <b>London, UK</b><br>Oct 2024 – Jun 2025       |
| <b>Thapathali Campus, Tribhuvan University</b><br><i>Adjunct Lecturer (Contract-based)</i> <ul style="list-style-type: none"><li>Lectures and labs for <i>A Practical Approach to Deep Learning</i>.</li><li>Supervised MS projects related to deep learning.</li></ul>                                                                                                                        | <b>Kathmandu, Nepal</b><br>Oct 2022 – May 2023 |
| <b>North Star Developer's Village</b><br><i>Engineering Manager</i> <ul style="list-style-type: none"><li>ML face verification; pose-based exam cheating prevention; image/audio annotation; ML training platform.</li><li>Semantic essay ranking; feedback bot; topic modelling/analysis; ML models exposed via APIs.</li></ul>                                                               | <b>Lalitpur, Nepal</b><br>Apr 2019 – May 2023  |
| <b>Leapfrog Technology</b><br><i>AI Intern</i> <ul style="list-style-type: none"><li>Predictive models for time-series weather and pollution forecasting (ARIMA and windowing-based).</li><li>Model deployment and web-based interactive interface.</li></ul>                                                                                                                                  | <b>Kathmandu, Nepal</b><br>Oct 2018 – Mar 2019 |
| <b>Department of Electronics &amp; Computer Engineering (DOECE)</b><br><i>Student Research Assistant for UGC Research Grant Project</i> <ul style="list-style-type: none"><li>Assisted Prof. Dr. Nanda Bikram Adhikari on algorithm/tool research &amp; implementation.</li><li>Sensor data storage with GPS traces &amp; data mining. Used RNN, RapidMiner, Hadoop, WebSocket, etc.</li></ul> | <b>Lalitpur, Nepal</b><br>Jan 2018 – Sep 2018  |

## Education

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| <b>Thapathali Campus, Tribhuvan University</b><br><i>Master's Degree in Informatics &amp; Intelligent Systems Engineering</i><br><i>Full Scholarship for Academic Merit, Top of Class</i><br><b>Relevant coursework:</b> Natural Language Processing; Computer Vision; Applied Machine Learning; Deep Learning; Computational Intelligence; Information Theory; Information Visualization; Optimization. | <b>Kathmandu, Nepal</b><br>Oct 2020 – Oct 2022 |
| <b>Pulchowk Campus, Tribhuvan University</b><br><i>Bachelor's Degree in Computer Engineering</i><br><i>Full Scholarship for Academic Merit</i><br><b>Relevant coursework:</b> Artificial Intelligence; Probability & Statistics; Data Mining; Big Data; Theory of Computation.                                                                                                                           | <b>Lalitpur, Nepal</b><br>Oct 2015 – Oct 2019  |

## Technical Capabilities

**Programming:** Python, Django, Android (Java/Kotlin), SQL, C/C++.  
**Libraries:** PyTorch, TensorFlow/Keras, Pandas, NumPy, scikit-learn, SciPy, Matplotlib.  
**Languages:** English (Fluent), Nepali (Native), Hindi (Advanced).

## Academic Projects

**MS Thesis:** AI-Based Soccer Game Summarization: From Video Highlights to Dynamic Text Summaries.  
Best thesis in the department (by scores). In collaboration with the Department of Holistic Systems, **SimulaMet**.  
**MS Project:** Deep Learning-based Pose Estimation for Dystonia Score Prediction. In collaboration with **NAAMII**.  
**Bachelor's Thesis:** Assessment of Spatiotemporal Weather and Pollution Proxies with RNN and ARIMA.  
Best Project Award by Department.  
**Bachelor's Project:** Human Facial Landmarks Detection using Multi-Layer Perceptron.

## Selected Publications

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- 2022:** Soccer Game Summarization using Audio Commentary, Metadata, and Captions. NarSUM, ACM Multimedia.
- 2022:** Assisting Soccer Game Summarization via Audio Intensity Analysis of Game Highlights. IOE Graduate Conference.
- 2020:** Near Real-Time Mobile Profiling and Modeling of Fine-Scale Environmental Proxies Along Major Road Lines of Nepal. Springer, Cham.
- 2019:** Sentence Ranking and Answer Pinpointing in Online Discussion Forums Utilizing User-generated Metrics and Highlights. NaSCoIT.
- 2018:** Real Time-Based Smart Traffic Light System with its Simulation using 8051 Microcontroller. KEC Conference.

## Awards & Certifications

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- NAAMII Winter School:** Full scholarship for participation. Topics: Deep Learning; Computer Vision; Statistics & Algorithms; Data Science; Optimization.
- Hult Prize TU Winner (2017):** Winning team "AEGIS". Presented a sustainable idea to empower refugees in Nepal using an e-commerce platform; presented at Hult Business School, Dubai.
- Mentor:** Global Venture Bootcamp, Asian Development Bank, Philippines — enhancing social ventures with emerging technologies.
- Participant:** IEEE AI & Big Data Workshop 2018; led multiple IEEE Student Chapter activities on campus.
- Participant:** UNICEF 1000 Golden Hours Code Camp (2017) — Best Prototype Award for mobile app idea.

## References

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- Prof. Pål Halvorsen:** Chief Research Scientist, SimulaMet; Professor, OsloMet; Adjunct Professor, UiO. [paalh@simula.uio.no](mailto:paalh@simula.uio.no) / [paalh@ifi.uio.no](mailto:paalh@ifi.uio.no); +47 97080007.
- Dr. Cise Midoglu:** Postdoctoral Fellow, Department of Holistic Systems (HOST), SimulaMet. [cise@simula.no](mailto:cise@simula.no); +47 92287195.
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